

KANEHOE MCAS, HI, USA

WMO: 911760

Lat: 21.450N

Lon: 157.768W

Elev: 24

StdP: 14.68

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 22519

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	63.8	65.8	53.1	60.0	70.5	54.8	63.9	71.1	23.7	72.3	20.7	71.7	2.0	180	0.341

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	7.8	85.8	75.3	84.5	74.7	83.8	74.5	77.6	82.2	76.8	81.7	76.1	81.3	10.2	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.2	136.9	80.3	75.2	132.1	80.1	74.2	127.9	79.7	41.0	82.9	40.1	81.6	39.4	81.3	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.7	16.8	15.6	DB	59.0	87.7	6.3	2.8	54.5	89.8	50.8	91.4	47.2	93.0	42.7	95.1
			WB	55.8	79.7	2.5	2.4	54.0	81.4	52.5	82.7	51.1	84.1	49.3	85.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	76.6	73.3	73.2	73.7	75.0	76.5	78.1	78.7	79.7	79.9	79.0	76.8	74.7
	DBStd	3.12	2.41	2.27	2.23	2.07	2.04	1.46	1.82	1.59	1.58	1.96	1.97	2.11
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9698	723	649	736	749	821	843	891	921	897	898	805	765
	CDD65	4223	258	229	271	299	356	393	426	456	447	433	355	300
	CDH74	26069	780	597	790	1102	1918	2755	3343	4039	3982	3550	2097	1116
	CDH80	2897	20	15	28	41	125	266	432	668	678	456	140	28
(14) Wind	WSAvg	8.4	7.3	8.1	8.5	9.0	8.1	9.1	9.4	8.9	8.1	7.9	8.1	7.9
(15) Precipitation	PrecAvg	32.6	5.2	2.9	3.1	3.7	2.1	1.2	1.8	1.5	1.7	2.8	3.6	4.3
	PrecMax	67.2	13.9	10.9	9.3	22.2	7.0	3.9	4.0	3.7	4.5	10.2	14.2	15.7
	PrecMin	12.8	0.4	0.2	0.2	0.6	0.0	0.4	0.5	0.4	0.3	0.7	0.7	0.4
	PrecStd	11.5	3.9	2.4	2.6	4.2	1.9	0.7	1.0	0.9	1.0	2.3	3.0	3.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.0	81.9	82.2	82.8	84.1	84.5	85.4	87.0	87.6	86.3	84.4	82.3
		MCWB	70.7	70.9	71.3	72.4	73.9	74.1	75.0	75.7	76.3	75.4	74.3	72.1
	2%	DB	80.4	80.0	80.9	81.2	82.3	83.4	84.2	85.3	85.9	84.4	82.5	81.0
		MCWB	70.3	70.0	70.4	71.3	72.9	73.7	74.6	75.1	75.6	74.9	73.4	71.6
	5%	DB	79.1	78.8	79.3	80.0	81.3	82.3	83.2	84.2	84.3	83.5	81.5	79.5
		MCWB	70.1	69.7	70.0	70.9	72.2	73.0	74.3	74.7	75.0	74.6	72.8	70.8
	10%	DB	77.9	77.4	78.1	79.0	80.6	81.5	82.3	83.1	83.3	82.3	80.7	78.8
		MCWB	69.6	69.2	69.7	70.5	71.8	72.6	73.7	74.3	74.5	73.9	72.4	70.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.2	74.4	74.3	74.9	75.6	76.3	78.2	78.4	79.1	78.2	77.0	75.1
		MCDB	77.8	77.9	78.9	79.3	80.9	81.5	81.5	82.7	83.6	82.5	81.5	78.6
	2%	WB	73.4	72.8	73.1	73.4	74.6	75.3	77.0	77.3	77.4	76.7	75.4	73.8
		MCDB	76.8	77.0	77.6	78.3	80.0	80.8	80.9	82.2	82.7	81.8	80.3	78.1
	5%	WB	72.3	71.5	72.0	72.6	73.7	74.5	76.0	76.3	76.4	75.8	74.5	73.0
		MCDB	76.3	76.0	76.6	77.7	79.1	80.3	80.7	81.7	82.1	81.3	79.5	77.6
	10%	WB	71.2	70.6	71.1	71.6	73.0	73.7	75.2	75.5	75.5	75.1	73.6	72.1
		MCDB	75.8	75.5	75.9	77.0	78.6	79.8	80.4	81.2	81.5	80.8	78.7	76.8
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	8.5	8.4	7.6	7.8	7.5	7.7	7.8	7.8	7.6	7.6	8.0
		MCDBR	10.7	10.5	10.1	8.8	8.6	8.1	8.5	8.8	8.9	8.7	8.9	9.2
	5% WB	MCWBR	5.9	5.7	5.3	4.6	4.4	4.2	4.3	4.3	4.2	4.4	4.8	5.1
		MCWBR	9.5	8.9	9.2	8.1	8.2	7.5	7.6	8.0	8.2	8.3	8.5	8.4
(40) Clear-Sky Solar Irradiance	taub	0.365	0.373	0.381	0.391	0.383	0.370	0.369	0.368	0.361	0.379	0.378	0.360	(40)
		2.567	2.533	2.521	2.480	2.532	2.575	2.589	2.593	2.616	2.568	2.562	2.602	(41)
	Ebn at Noon	288	292	293	290	289	290	290	292	295	286	282	286	(42)
		30	32	34	36	34	32	31	31	30	31	30	28	(43)
(44) All-Sky Solar Radiation	RadAvg	1387	1564	1796	2027	2178	2285	2241	2150	2012	1688	1421	1283	(44)
	RadStd	59	110	187	113	121	63	48	85	61	51	36	91	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.67	N/A	N/A	+1.08	+0.70	+0.81	N/A	N/A	+250	+252
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+122	+122

Nomenclature: See separate page